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1.5 Trigonometry

Trigonometric Ratios

For a right-angled triangle with angle θ, opposite side O, adjacent side A, and hypotenuse H:

$$\sin\theta = \frac{O}{H}, \quad \cos\theta = \frac{A}{H}, \quad \tan\theta = \frac{O}{A} = \frac{\sin\theta}{\cos\theta}$$

Exact Trigonometric Values

θ	$0°$	$30°$	$45°$	$60°$	$90°$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undef.

θ (radians)	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0

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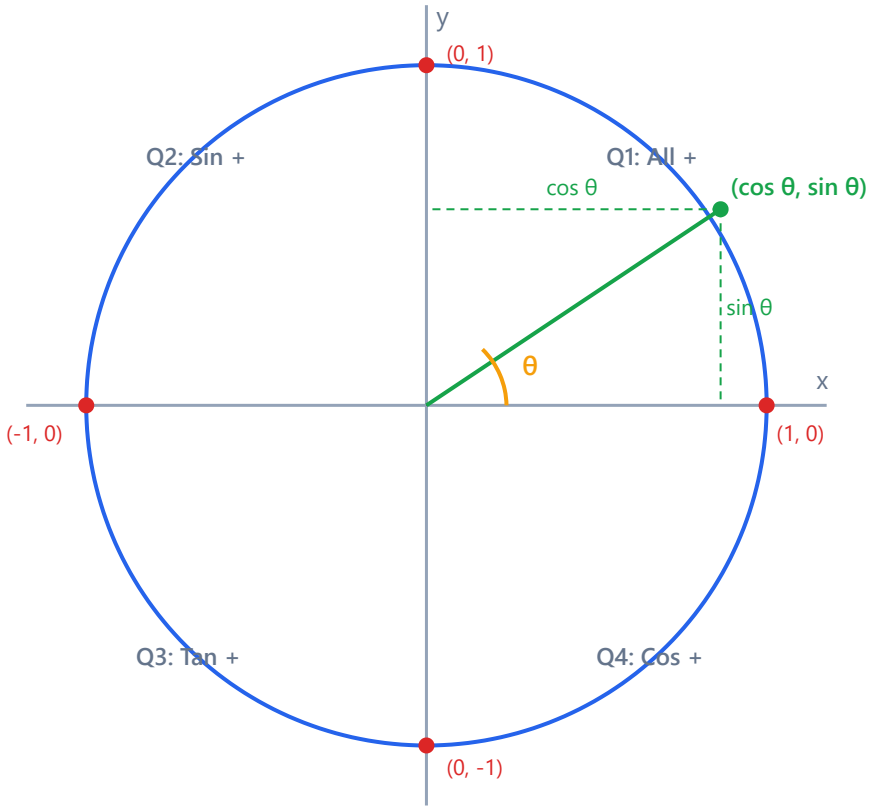
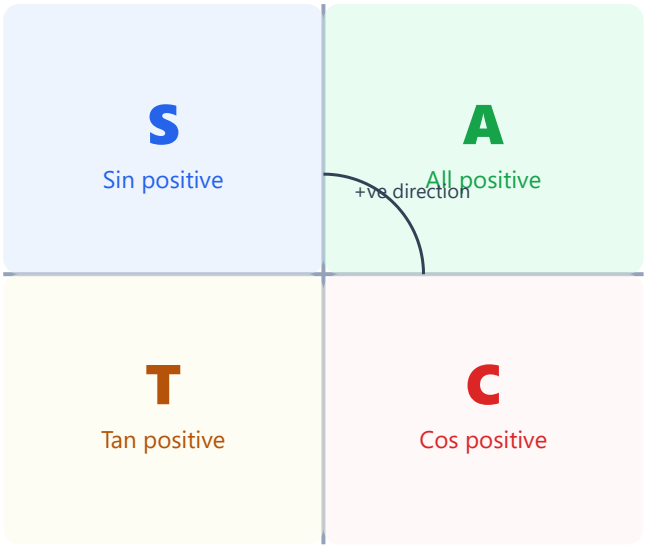
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undef.

The Unit Circle

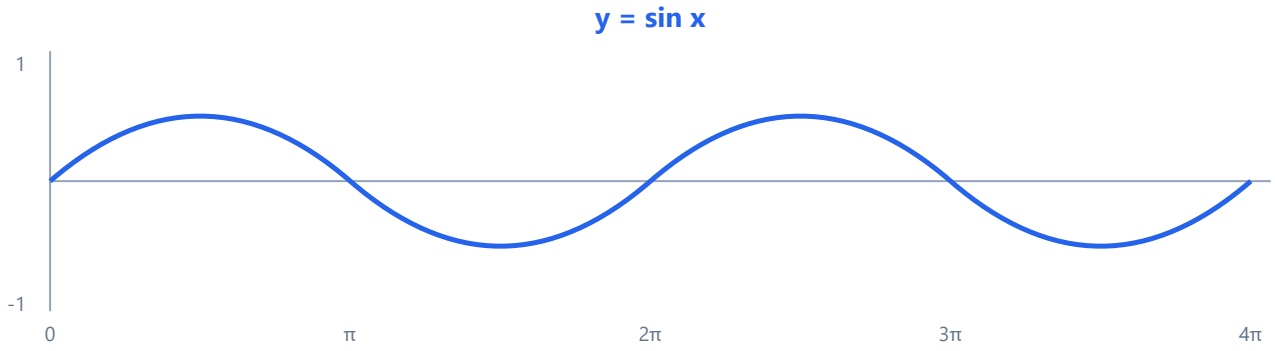


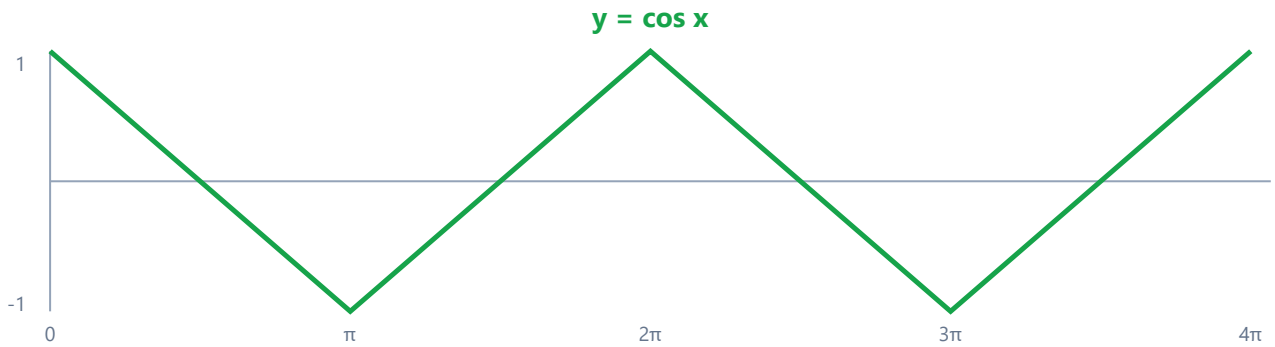
CAST Diagram

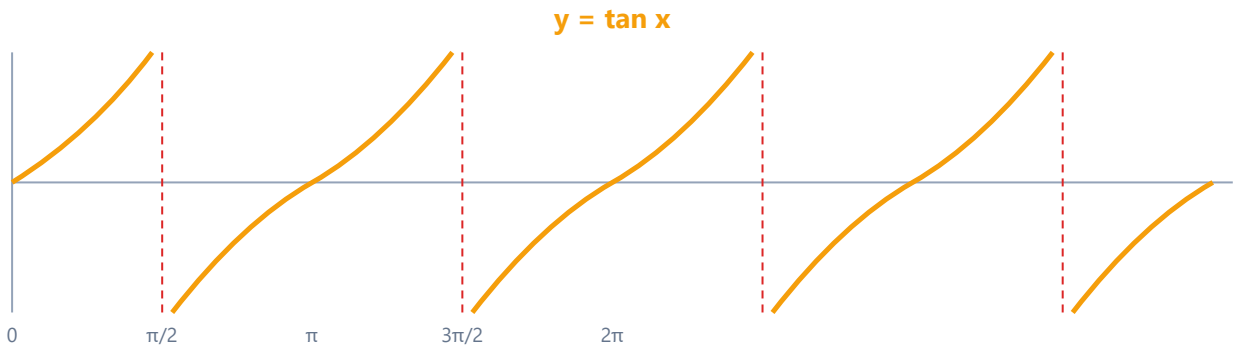


Remember: All Students Take Calculus

Graphs of Trigonometric Functions







Function	Period	Range	Key Properties
$\sin x$	2π (360 deg)	$[-1, 1]$	Odd function: $\sin(-x) = -\sin x$
$\cos x$	2π (360 deg)	$[-1, 1]$	Even function: $\cos(-x) = \cos x$
$\tan x$	π (180 deg)	$(-\infty, \infty)$	Odd function, asymptotes at $x = \frac{\pi}{2} + n\pi$

Reciprocal Trigonometric Functions

$$\sec\theta = \frac{1}{\cos\theta}, \quad \csc\theta = \frac{1}{\sin\theta}, \quad \cot\theta = \frac{1}{\tan\theta} = \frac{\cos\theta}{\sin\theta}$$

Trigonometric Identities

Pythagorean Identities

$$\sin^2\theta + \cos^2\theta = 1$$

$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2\theta = \csc^2\theta$$

Addition Formulae

$$\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$$

$$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}$$

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Double Angle Formulae

$$\sin 2A = 2\sin A\cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$$

$$\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$$

Identity Summary Table

Identity	Formula
Quotient	$\tan\theta = \dfrac{\sin\theta}{\cos\theta}$
Pythagorean 1	$\sin^2\theta + \cos^2\theta = 1$
Pythagorean 2	$1 + \tan^2\theta = \sec^2\theta$
Pythagorean 3	$1 + \cot^2\theta = \csc^2\theta$
Double angle (sin)	$\sin 2A = 2\sin A\cos A$
Double angle (cos)	$\cos 2A = \cos^2 A - \sin^2 A$
Double angle (tan)	$\tan 2A = \dfrac{2\tan A}{1 - \tan^2 A}$
Half-angle from cos 2A	$\cos^2 A = \dfrac{1 + \cos 2A}{2}$, $\sin^2 A = \dfrac{1 - \cos 2A}{2}$

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Example: Prove that $\dfrac{\sin 2\theta}{1+\cos 2\theta} = \tan\theta$.

LHS: $\dfrac{2\sin\theta\cos\theta}{1+2\cos^2\theta-1} = \dfrac{2\sin\theta\cos\theta}{2\cos^2\theta} = \dfrac{\sin\theta}{\cos\theta} = \tan\theta$ = RHS. $\square$

The $R\sin(x+\alpha)$ and $R\cos(x+\alpha)$ Form

Any expression $a\sin x + b\cos x$ can be written as:

$$a\sin x + b\cos x = R\sin(x+\alpha)$$

where $R = \sqrt{a^2+b^2}$ and $\tan\alpha = \dfrac{b}{a}$.

Similarly, $a\cos x + b\sin x = R\cos(x-\alpha)$ where $\tan\alpha = \dfrac{b}{a}$.

Example: Write $3\sin x + 4\cos x$ in the form $R\sin(x+\alpha)$.

$R = \sqrt{3^2+4^2} = \sqrt{25} = 5$

$\tan\alpha = \dfrac{4}{3} \Rightarrow \alpha = \arctan\left(\dfrac{4}{3}\right) \approx 53.13^\circ$

$3\sin x + 4\cos x = 5\sin(x+53.13^\circ)$

Maximum value is 5 (when $\sin(x+53.13^\circ) = 1$).

Minimum value is -5 (when $\sin(x+53.13^\circ) = -1$).

Exam Tip: The $R\sin(x+\alpha)$ form is used to find the maximum/minimum of a trig expression and to solve equations of the form $a\sin x + b\cos x = c$. The max value is always R and the min is $-R$.

Small Angle Approximations

When θ is small and measured in **radians**:

$$\sin\theta \approx \theta, \quad \cos\theta \approx 1 - \frac{\theta^2}{2}, \quad \tan\theta \approx \theta$$

Example: Find an approximation for $\dfrac{1-\cos 2\theta}{\theta \sin\theta}$ when θ is small.

Using small angle approximations:

Numerator: $1 - \cos 2\theta \approx 1 - \left(1 - \dfrac{(2\theta)^2}{2}\right) = 2\theta^2$

Denominator: $\theta \sin\theta \approx \theta \cdot \theta = \theta^2$

$$\frac{1-\cos 2\theta}{\theta\sin\theta} \approx \frac{2\theta^2}{\theta^2} = 2$$

Inverse Trigonometric Functions

- $\arcsin x$ (or $\sin^{-1} x$): domain $[-1, 1]$, range $\left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$
- $\arccos x$ (or $\cos^{-1} x$): domain $[-1, 1]$, range $[0, \pi]$
- $\arctan x$ (or $\tan^{-1} x$): domain $(-\infty, \infty)$, range $\left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$

Solving Trigonometric Equations

General Strategy

1. Rearrange to isolate a single trig function if possible.
2. Find the **principal value** using inverse trig functions.
3. Use the CAST diagram or symmetry to find **all solutions** in the given range.
4. If the equation involves 2θ or 3θ, adjust the range accordingly before finding solutions, then divide at the end.

Example: Solve $2\sin^2 x + \sin x - 1 = 0$ for $0 \leq x < 2\pi$.

This is a quadratic in $\sin x$. Let $u = \sin x$:

$2u^2 + u - 1 = 0 \Rightarrow (2u-1)(u+1) = 0$

$u = \frac{1}{2}$ or $u = -1$.

Case 1: $\sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}$ or $x = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.

Case 2: $\sin x = -1 \Rightarrow x = \frac{3\pi}{2}$.

Solutions: $x = \dfrac{\pi}{6}, \dfrac{5\pi}{6}, \dfrac{3\pi}{2}$.

Example: Solve $\cos 2\theta = 0.5$ for $0^\circ \leq \theta \leq 360^\circ$.

Since we have 2θ, adjust range: $0^\circ \leq 2\theta \leq 720^\circ$.

$\cos 2\theta = 0.5$: principal value $2\theta = 60^\circ$.

From the CAST diagram / cosine graph, solutions for 2θ:

$2\theta = 60^\circ, 300^\circ, 420^\circ, 660^\circ$

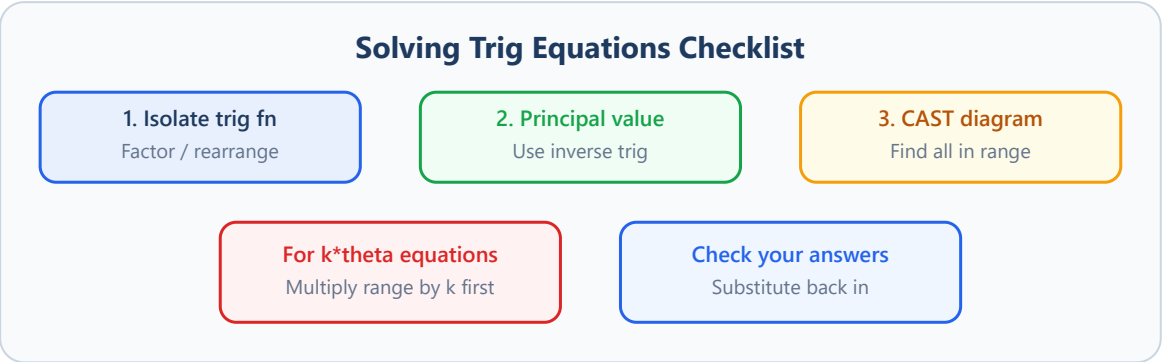
Divide by 2: $\theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$.

Example: Solve $3\sin x = 2\cos x$ for $0^\circ \leq x < 360^\circ$.

Divide both sides by $\cos x$: $3\tan x = 2 \Rightarrow \tan x = \frac{2}{3}$.

Principal value: $x = \arctan\left(\frac{2}{3}\right) = 33.69°$.

Tan is positive in Q1 and Q3: $x = 33.69°$ or $x = 180° + 33.69° = 213.69°$.



Key exam strategies for trigonometry:

- Memorise the exact values table -- these come up constantly.
- For identities, always work from the more complicated side towards the simpler one.
- When you see $\sin^2$ or $\cos^2$, think about the Pythagorean identity.
- For equations with mixed trig functions, try to express everything in terms of one function.
- Always check the required range and units (degrees vs radians).
- For $R\sin(x + \alpha)$ questions, expand using the addition formula and compare coefficients to find R and α.
- Small angle approximations only work in **radians**.